JINFENG XU

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RESEARCH AREA AND INTERESTS

3D Computer Vision · Scene Understanding · Open-World Learning · Semantic Scene Completion

EXPERIENCE

Hunan University *Jul. 2026 – Now*
Assistant Professor at the Institute for Artificial Intelligence and Advanced Manufacturing. *Changsha, China*

EDUCATION

Huazhong University of Science and Technology *Sep. 2020 – Jun. 2026*
Ph.D. student in Computer Science and Technology, Supervisor: Xianzhi Li. *Wuhan, China*

Shandong University *Sep. 2016 – Jun. 2020*
Bachelor of Engineering in Automation, Rank: 13/131 (10%). *Shandong, China*

PUBLISHED PAPER

[1] **Jinfeng Xu**, Xianzhi Li, Yuan Tang, Xu Han, Qiao Yu, Yixue Hao, Long Hu, Min Chen. SASep: Saliency-Aware Structured Separation of Geometry and Feature for Open Set Learning on Point Clouds. In **CVPR** 2025.
⋆ *Addressed the underexplored problem of 3D open-set recognition by introducing a saliency-aware part-based framework that advances unknown-class discovery. [Code]*

[2] **Jinfeng Xu**, Siyuan Yang, Xianzhi Li, Yuan Tang, Yixue Hao, Long Hu, Min Chen. PDF: A Probability-Driven Framework for Open World 3D Point Cloud Semantic Segmentation. In **CVPR** 2024.
⋆ *Filled the gap in 3D open-world segmentation with a probability-driven framework that unifies uncertainty estimation and continual learning by pseudo-labeling and knowledge distillation, significantly outperforming the existing approaches. [Code]*

[3] **Jinfeng Xu**, Xianzhi Li, Yuan Tang, Qiao Yu, Yixue Hao, Long Hu, Min Chen. CasFusionNet: A Cascaded Network for Point Cloud Semantic Scene Completion by Dense Feature Fusion. In **AAAI** 2023.
⋆ *Proposed a cascaded dense-fusion network for point-cloud semantic scene completion and contributed two new benchmark datasets, enabling comprehensive evaluation of SSC methods. [Code]*

[4] Yuan Tang, Yunlong Yu, Xianzhi Li, Rui Wang, **Jinfeng Xu**, Qiao Yu, Yixue Hao, Long Hu, Min Chen. Rethinking Point Cloud Representation Learning for Freeing Transformer to Perceive Local. **IEEE TMM** 2026.

[5] Xu Han, Yuan Tang, **Jinfeng Xu**, Xianzhi Li. MoST: Efficient Monarch Sparse Tuning for 3D Representation Learning. In **CVPR** 2025.

[6] Qiao Yu, Xianzhi Li, Yuan Tang, Xu Han, **Jinfeng Xu**, Long Hu, Min Chen. PointDreamer: Zero-Shot 3D Textured Mesh Reconstruction From Colored Point Cloud. **IEEE TVCG** 2025.

[7] Yuan Tang, Xu Han, Xianzhi Li, Qiao Yu, **Jinfeng Xu**, Yixue Hao, Long Hu, Min Chen. More Text, Less Point: Towards 3D Data-Efficient Point-Language Understanding. In **AAAI** 2025.

[8] Qiao Yu, Xianzhi Li, Yuan Tang, **Jinfeng Xu**, Long Hu, Yixue Hao, Min Chen. JIMR: Joint Semantic and Geometry Learning for Point Scene Instance Mesh Reconstruction. **IEEE TVCG** 2024.

[9] Yuan Tang, Xianzhi Li, **Jinfeng Xu**, Qiao Yu, Long Hu, Yixue Hao, Min Chen. Point-LGMask: Local and Global Contexts Embedding for Point Cloud Pre-Training With Multi-Ratio Masking. **IEEE TMM** 2023.

PROJECTS

3D Reconstruction and Semantic Understanding with Limited Annotations | *NSFC (China)* *Aug. 2022 – Dec. 2025*

- Participated in a national research project (China), focusing on 3D scene reconstruction and semantic understanding under weakly-labeled data conditions.
- Contributed to algorithm development for semantic scene completion, open-world 3D segmentation, and open-set recognition, leading to three first author publications (AAAI'23, CVPR'24, CVPR'25).

Semantic-Prior-Driven Building Facade Modeling | *Invention Patent* *Oct. 2023 – Dec. 2023*

- Proposed a semantic segmentation–based pipeline with prior-guided optimization for facade element modeling.
- Resulted in a Chinese patent application (202410270767X, under examination).

Large-Scale Crowd Monitoring and Management Platform with UAVs | *Collaboration with KAUST* *Feb. 2021 – May. 2022*

- Designed and implemented a customized image transmission protocol using socket data frames and JSON encapsulation for low-latency and reliable video streaming.
- Integrated compression and concurrency control into a ROS-based UAV system, ensuring stable video streaming and enabling real-time target tracking for large-scale crowd monitoring.

Vision-Guided Robotic Manipulation System | *Research Collaboration* *Feb. 2020 – Dec. 2021*

- Developed an autonomous robotic pick-and-place pipeline integrating RGB-D perception, calibration, and robotic control for shelf-based item retrieval.
- Implemented hand–eye calibration to map 3D object poses from depth cameras to the robotic arm coordinate system, enabling precise grasp execution.

Honors and Awards

- 2024 Huawei Scholarship.
- 2021-2024 Ph.D. First Class Academic Scholarship.
- 2019 National Special Prize, National Engineering Training Competition (Robotics), Ministry of Education.
- 2018 National Second Prize, National Intelligent Vehicle Competition (NXP Cup), Ministry of Education.